

MIAE 221: Materials Science for Engineers

Chapters 12 & 13: Structures, Properties & Processing of Ceramics (Expanded & Condensed Guide)

Concordia University · Gina Cody School of Engineering

Textbook:

Materials Science and Engineering: An Introduction

(10th Ed., Callister & Rethwisch)

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(Syllabus Week 9 · Sections 12.3–12.11 & 13.11)

1. 🌀 30-Second Core Intuition & Plain-English Concept

Ceramics are compounds of metallic and nonmetallic elements held by strong **ionic and covalent bonds**.

Because ions have fixed positive and negative charges, like-charges fiercely repel if atomic planes attempt to slide. Therefore, **dislocation motion is virtually impossible** at room temperature. Ceramics cannot plastically deform; under tension, microscopic surface microcracks concentrate stress and propagate instantaneously at the speed of sound (**brittle catastrophic fracture**).

Real-World Analogy

: Trying to slide a row of alternating north-south magnets past another row. As soon as you shift them half a notch, all north poles align with north poles, and the entire structure blasts violently apart.

2. ⚙️ High-Yield Mathematical Engine & Ceramic Geometry

1. Cation-to-Anion Radius Ratio (r_C/r_A) Coordination Table

Cations are smaller than anions ($r_C < r_A$). Stable crystal structures require cations to maximize contact with surrounding anions without allowing anions to overlap:

Radius Ratio (r_C/r_A)	Coordination Number (CN)	Coordination Geometry	Archetypal Ceramic Crystal Structure
< 0.155	2	Linear	Inert gas configurations
$0.155 - 0.225$	3	Triangular planar	$\text{extB}_2\text{extO}_3$
$0.225 - 0.414$	4	Tetrahedral	Zinc Blende (extZnS), extSiO_4^{4-}
$0.414 - 0.732$	6	Octahedral	Rock Salt (extNaCl , extMgO , extFeO)
$0.732 - 1.000$	8	Cubic	Cesium Chloride (extCsCl)
> 1.000	12	Close-packed	Rare intermetallic ceramics

2. Defect Chemistry in Ceramics

To preserve **electroneutrality** (charge neutrality):

- **Frenkel Defect:** A cation vacates its normal lattice site and squeezes into a nearby interstitial position (cation vacancy + cation interstitial; no net charge change).
- **Schottky Defect:** A stoichiometric pair consisting of **one cation vacancy AND one anion vacancy** (e.g., in *extNaCl*, one *extNa*⁺ missing and one *extCl*⁻ missing; net charge remains zero).

3. Flexural Strength via Three-Point Bending (σ_{fs})

Because ceramics break prematurely in tension grips, tensile strength is measured via transverse bending:

$$\sigma_{fs} = \frac{3F_f L}{2bd^2} \quad (\text{Rectangular cross-section: width } b, \text{ depth } d, \text{ support span } L)$$
$$\sigma_{fs} = \frac{F_f L}{\pi R^3} \quad (\text{Circular cross-section of radius } R)$$

3. 🖼️ Textbook Reference Diagram & Visual Pedagogical Analysis

$$\overline{AO} = r_A + r_C$$

Furthermore, the side length ratio $\overline{AP}/\overline{AO}$ is a function of the angle α as

$$\frac{\overline{AP}}{\overline{AO}} = \cos \alpha$$

magnitude of α is 30° , because line $\overline{AO}$ bisects the 60° angle BAC . Thus,

$$\frac{\overline{AP}}{\overline{AO}} = \frac{r_A}{r_A + r_C} = \cos 30^\circ = \frac{\sqrt{3}}{2}$$

ing for the cation–anion radius ratio,

$$\frac{r_C}{r_A} = \frac{1 - \sqrt{3}/2}{\sqrt{3}/2} = 0.155$$

AX-Type Crystal Structures

Some of the common ceramic materials are those in which there are equal numbers of cations and anions. These are often referred to as AX compounds, where A denotes the cation and X the anion. There are several different crystal structures for AX compounds; each is typically named after a common material that assumes the particular structure.

Figure 12.2: Rock Salt (*extNaCl*) unit cell. FCC anion packing with cations occupying all octahedral interstices.

oxides do not form polyhedral networks. Rather, their cations are incorporated within and modify the SiO_4^{4-} network; for this reason, these oxide additives are termed *network modifiers*. For example, Figure 12.11 is a schematic representation of the structure of a sodium–silicate glass. Still other oxides, such as TiO_2 and Al_2O_3 although not network formers, substitute for silicon and become part of and stabilize the network; these are called *intermediates*. From a practical perspective, the addition of these modifiers and intermediates lowers the melting point and viscosity of a glass and makes it easier to form at lower temperatures (Section 13.11).

The Silicates

For the various silicate minerals, one, two, or three of the corner oxygen atoms of the SiO_4^{4-} tetrahedra are shared by other tetrahedra to form some rather complex structures. Some of these, represented in Figure 12.12, have formulas SiO_4^{4-} , $\text{Si}_2\text{O}_7^{6-}$, $\text{Si}_3\text{O}_9^{6-}$ and so on; single-chain structures are also possible, as in Figure 12.12e. Positively charged cations such as Ca^{2+} , Mg^{2+} , and Al^{3+} serve two roles: First, they compensate the negative charges from the SiO_4^{4-} units so that charge neutrality is achieved; second, these cations ionically bond the SiO_4^{4-} tetrahedra together.

Figure 12.10: 2D representation of noncrystalline silica glass with network modifier cations (*ext* Na^+ , *ext* Ca^{2+}) disrupting the bridging oxygen network.

raised, the magnitude of the bonding is diminished, the sliding motion or flow of the atoms or ions is facilitated, and subsequently there is an attendant decrease in viscosity. A discussion of the temperature dependence of viscosity for glasses is deferred to Section 13.11.

Figure 12.32 Representation of the viscous flow of a liquid or fluid glass in response to an applied shear force.

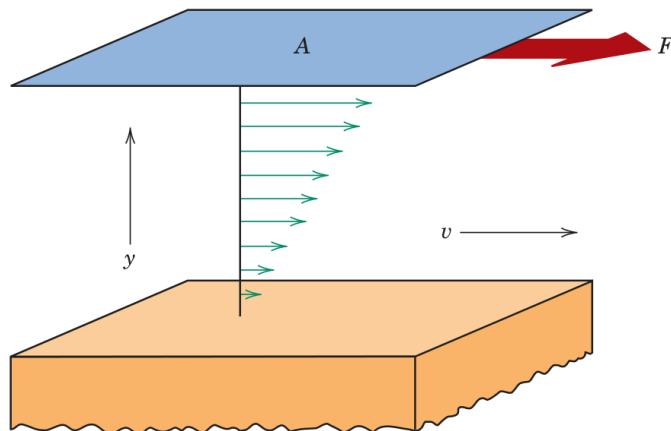


Figure 12.32: Three-point bending fixture for measuring flexural strength of brittle ceramics.

In-Depth Visual Breakdown:

- **Rock Salt Structure (Figure 12.2):** Anions (*ext* Cl^-) form an FCC unit cell lattice. Octahedral interstitial sites are located at the center of the unit cell and at the midpoints of all 12 cube edges. Cations (*ext* Na^+) fill all octahedral sites ($n = 4$ formula units per cell).
- **Silicate Glass (Figure 12.10):** Pure crystalline silica (*ext* SiO_2) forms a rigid, ordered network of corner-sharing *ext* SiO_4^{4-} tetrahedra. Adding network modifiers like soda (*ext* Na_2extO) and lime (*ext* CaO) introduces non-bridging oxygen atoms, breaking up network connectivity, drastically lowering the melting point from 1710°extC to $\approx 1000^\circ\text{extC}$ to produce workable window glass.

4. 🏛️ Teacher Notes Cross-Reference & Concordia Exam Traps

- **Concordia Exam Focus:**
 - Predicting ceramic coordination number and crystal structure from ion radii ($r_{\text{extcation}}$ and r_{extanion}).
 - Calculating flexural strength σ_{fs} from 3-point bend test failure loads.
 - Explaining why ceramics have high compressive strength (10 $\times$ higher) but low tensile strength (Griffith flaw theory: microcracks open in tension but close harmlessly in compression).
- **Concordia Exam Traps:**
 - Forgetting that in the 3-point bending formula for rectangular bars ($\sigma_{fs} = \frac{3FL}{2bd^2}$), the depth d is **squared**, whereas width b is linear. Inverting b and d will produce massive numerical errors!

5. 💡 Master Exam Problem & Step-by-Step Solution Framework

Problem:

Magnesium oxide (extMgO) has $r_{\text{extMg}^{2+}} = 0.072 \text{ extnm}$ and $r_{\text{extO}^{2-}} = 0.140 \text{ extnm}$. (a) Predict the coordination number and crystal structure. (b) A rectangular bar of extMgO ($b = 10 \text{ extmm}$, $d = 5 \text{ extmm}$) tested in a 3-point bend fixture with span $L = 50 \text{ extmm}$ fractures at load $F_f = 2,500 \text{ extN}$. Compute its flexural strength.

- **Step 1: Compute Cation-Anion Radius Ratio**

$$\frac{r_{\text{Mg}^{2+}}}{r_{\text{O}^{2-}}} = \frac{0.072 \text{ nm}}{0.140 \text{ nm}} = 0.514$$

- **Step 2: Predict Coordination and Structure**

Since $0.414 < 0.514 < 0.732$, the coordination number is **6 (Octahedral)**. Because stoichiometry is 1 : 1 (extAX type), extMgO crystallizes in the **Rock Salt (extNaCl) crystal structure**.

- **Step 3: Calculate Flexural Strength σ_{fs}**

$$\sigma_{fs} = \frac{3F_f L}{2bd^2} = \frac{3(2,500 \text{ N})(0.050 \text{ m})}{2(0.010 \text{ m})(0.005 \text{ m})^2} = \frac{375}{2(0.010)(2.5 \times 10^{-5})} = \frac{375}{5.0 \times 10^{-7}} = 750 \times 10^6 \text{ Pa} = 750 \text{ MPa}$$
